

6.2.4 Carbon–carbon bond formation

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Extending carbon chain length	
(a) the use of C–C bond formation in synthesis to increase the length of a carbon chain (see also 6.1.1 d, 6.1.2 b)	
(b) formation of C–C≡N by reaction of:	
(i) haloalkanes with CN [−] and ethanol, including nucleophilic substitution mechanism (see also 4.2.2 c)	
(ii) carbonyl compounds with HCN, including nucleophilic addition mechanism (see also 6.1.2 b–c)	
(c) reaction of nitriles from (b) :	
(i) by reduction (e.g. with H ₂ /Ni) to form amines	
(ii) by acid hydrolysis to form carboxylic acids	

- (d)** formation of a substituted aromatic C–C by alkylation (using a haloalkane) and acylation (using an acyl chloride) in the presence of a halogen carrier (Friedel–Crafts reaction) **(see also 6.1.1 d).**